The Meaning of Compute

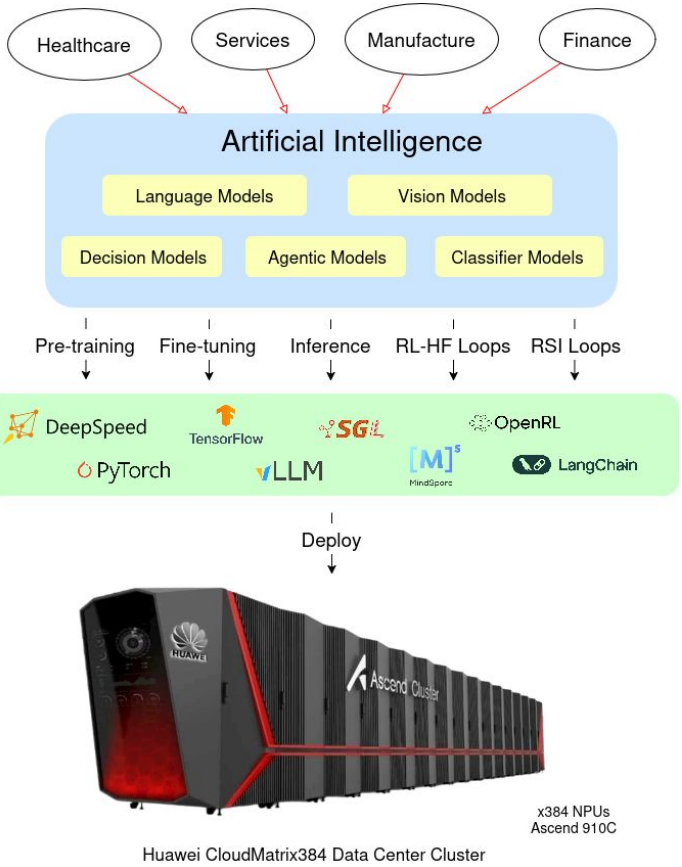
Formal approaches to exascale AI development



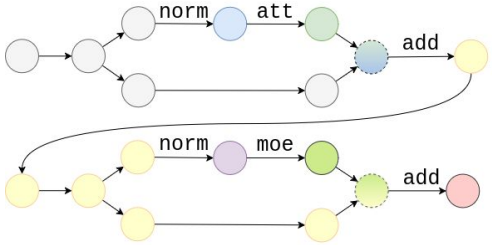
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AI requires compute power!

- Infrastructure for “heavy computing”
- Fast pace, constant adaptation
- Highest quality of code



AI computing is very tricky!

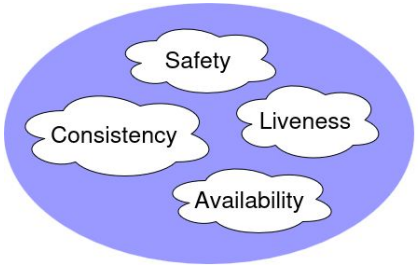


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- Data Parallelism
- Tensor Parallelism
- Pipeline Parallelism
- Expert Parallelism
- Sequence Parallelism
- ...

Reasoning about the properties of a program

Formal approaches are mostly applied to study critical programs

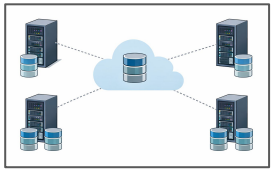


Motivate the creation of

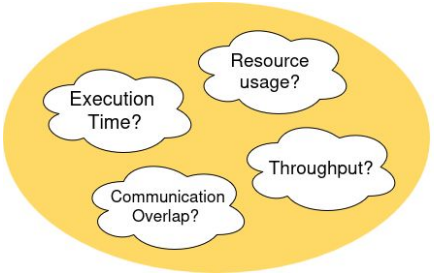
Type Systems	Program Synthesizers
Static Analyzers	Specification Languages
Theorem Provers	Model Checkers

Enable safety and reliability





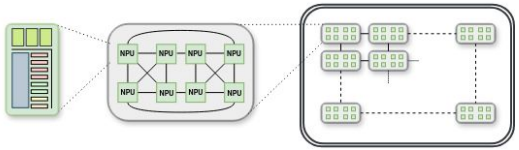
But AI workloads are programs that demand critical performance!



Motivate the creation of

?

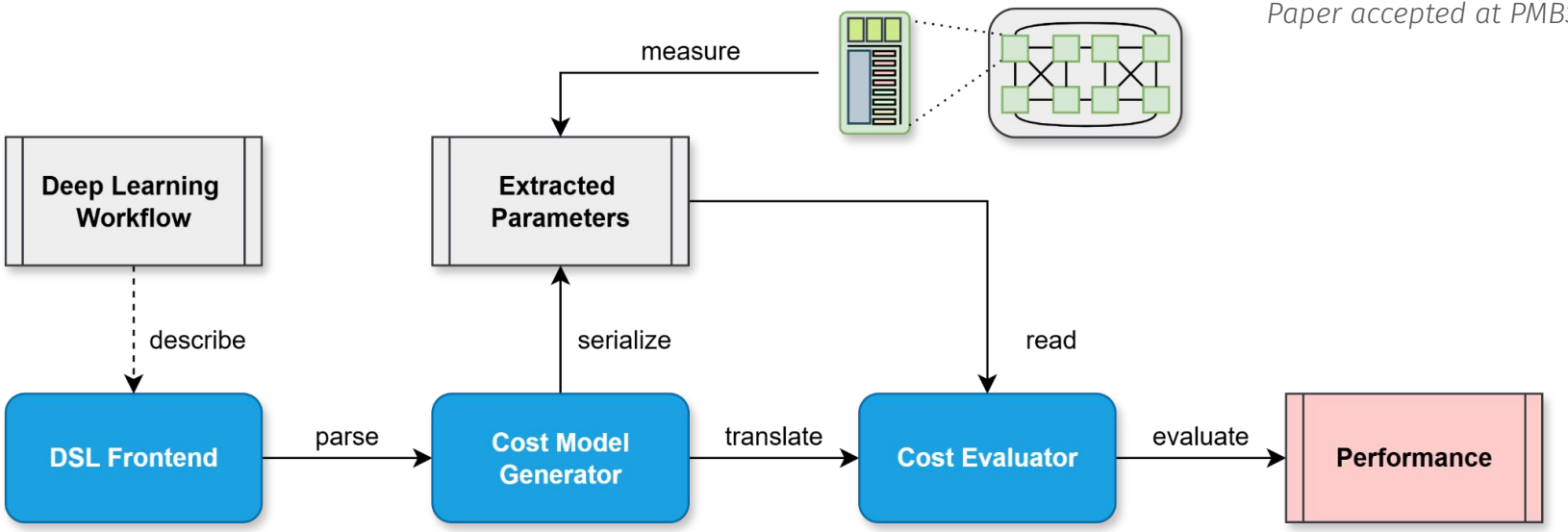
Enable guarantees of performance



AMAI: Algebraic Modelling of AI

*"Shift the focus: from **what** to compute, to **how** to compute"*

Paper accepted at PMBS '26



Let’s work together to push the frontier further!

